SHARED RECOVERY PROFILES, CALIBRATED CELL RESPONSES, AND SCALE-CONDITIONED LEARNING IN ARBITRARY-SCALE SUPER-RESOLUTION

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ABSTRACT

Arbitrary-scale super-resolution combines learned image features, continuous decoding and target-pixel-size conditioning. We identify the frequency range in which additional capacity helps, and distinguish the response to cell size from its value during learning. First, recovery profiles are closely shared by the tested decoder families, while stronger encoders, larger MLPs and longer training improve recovery mainly from input-normalised frequency $u \approx 0.75$ upward. This low-frequency saturation persists under a second, Gaussian-blur degradation. Second, fixed cell multipliers change the actual decoder-input step with scale. We control this step directly and separate output-change energy from its alignment with reconstruction error. A prospective full-image test finds 11 materially increasing adjacent-scale responses under family-wide correction, so declining fixed-multiplier cost does not establish universal response decay. Third, dimension-preserving constant-cell and shuffled-cell training test the contribution of correctly paired conditioning under a matched 200-epoch recipe. These training comparisons are in progress; this review draft makes no training-source claim. Together, these experiments connect frequency-localised capacity gains with controlled measurements of scale conditioning.

***Index Terms*—** Arbitrary-scale super-resolution, implicit representation, frequency recovery, scale conditioning

1. INTRODUCTION

Continuous-coordinate super-resolution decodes local image features at arbitrary output locations. LIIF [1] conditions this decoder on target pixel size, or *cell*; subsequent designs use local Fourier estimators [2], integrated positional encoding [3], implicit attention [4], neural operators [5], Gaussian splatting [6] or heat-field filtering [7]. Which parts of the recovered signal benefit from these choices, and what information does cell contribute?

Prior work links performance to model size and training [8] and studies spectral decoding's efficiency–accuracy trade-off [9]. HyRA analyses nonlinear compensation of low-pass phase distortion [10]; LIIF, integrated encoding and heat fields connect cell to edges and filtering [1, 3, 7]. We locate capacity gains and distinguish cell-induced output change, error cost and training benefit.

We show that (1) tested decoder forms share recovery profiles, with capacity gains concentrated near and above $u = 0.75$; (2) calibrated cell steps reject universal response decay while separating response from error alignment; and (3) dimension-preserving training controls test correctly paired conditioning. The training-source verdict remains pending. Table 2 retains original failures.

2. SHARED RECOVERY PROFILES AND CAPACITY

Models and measurement. LIIF, LTE and SC-INR (an LTE-derived experimental variant with analytic sinc cell weights) use the EDSR-baseline architecture [11], separately learned encoder weights and a common DIV2K [12] recipe: scales from $\mathcal{U}(1, 4)$, 1000 epochs and three seeds. GaussianSR uses an external 2000-epoch EDSR checkpoint and provides an additional comparison. Matched 200-epoch LIIF arms change the encoder to RDN [13] or SwinIR [14], reduce the MLP from 4×256 to 2×64, or remove cell. These comparisons match recipes; parameter counts and compute differ.

Let $r = \max(|f_x|, |f_y|)$ be frequency in cycles per output pixel and $u = 2sr$ its value relative to the input Nyquist limit at scale s. Recovery in band b is $R_b = 1 - V_b/E_b$, where error energy V_b and target energy E_b are pooled over images before taking the ratio. Original crop analyses use DIV2K 0869–0900, 240^2 centre crops, quarter-u bins, anti-aliased bicubic inputs and a four-pixel shave. Full-image recovery additionally covers Set14 [15], BSD100 [16] and Urban100 [17]. We evaluate $\times 2/3/4/6$; $\times 6$ lies outside the training scale range.

A shared recovery regime. Across $\times 2/3/4$, the three matched decoder families have a maximum scale-specific band spread of 0.010, attained at $\times 4$, in the registered $0.5 \leq u < 2$ comparison (paired image-bootstrap 95% interval $[0.007, 0.016]$ at that scale). With GaussianSR included, all 54 full-image dataset–scale–band comparisons stay below 0.019. Closely shared profiles extend across the tested decoder forms and image collections (Fig. 1a,b), although the stricter cross-scale-collapse prediction fails its registered tolerance gates.

Capacity improves the incompletely recovered bands. Longer training, stronger encoders and a larger MLP change recovery primarily from $u \approx 0.75$ upward (Fig. 1c). From epoch 100 to 1000, gains in $1 \leq u < 2$ range from 0.0245 to

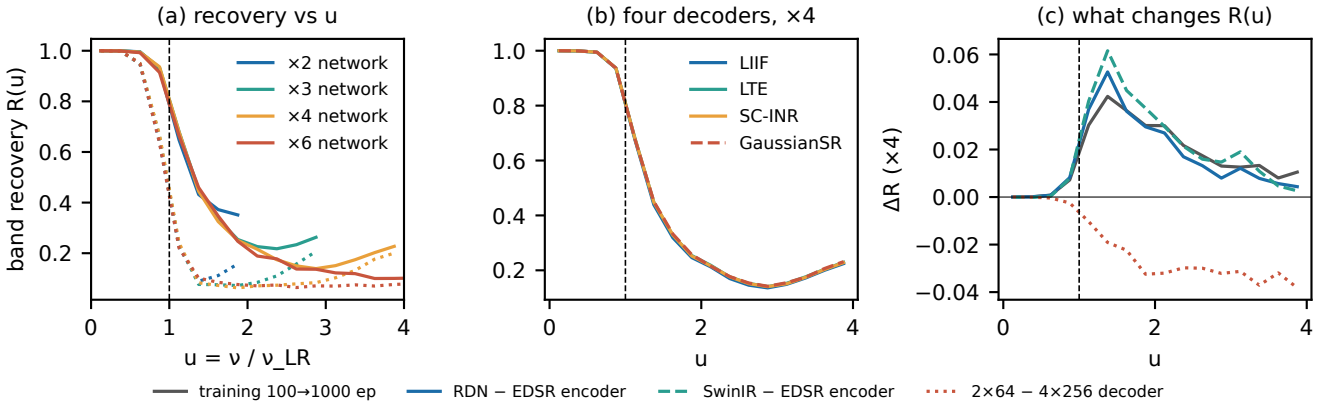


Fig. 1. Shared recovery profiles and frequency-localised capacity effects. (a) Three decoder families and three seeds at four scales; dotted curves are bicubic. (b) Decoder comparison at $\times 4$; GaussianSR is an external checkpoint with a different recipe. (c) Effects of training duration, encoder and MLP capacity. The plotted points represent frequency bands.

0.0464 across the three decoder families and $\times 2/3/4$, while changes below $u = 0.75$ are at most 0.0013. Matched RDN and smaller-MLP experiments replicate the high-frequency improvement and loss, respectively, with a second training seed. Decoder form and decoder capacity therefore have distinct empirical effects.

After retraining with Gaussian blur ($\sigma = 0.6s$ HR pixels) and non-antialiased bicubic resizing, RDN improves PSNR at all scales but changes recovery below $u = 0.75$ by at most 0.0006, failing the registered lower-band-gain prediction. This one-seed result extends low-frequency saturation to a second degradation. Table 1 and Fig. 3 show benchmark and visual effects.

3. CALIBRATING THE RESPONSE TO CELL

Equal multipliers imply unequal interventions. For output side length L, the implementations multiply the supplied cell $2/L$ by the LR feature-grid size L/s. The decoder receives $c = 2/s$ on each axis. Multiplying cell by m therefore gives the per-axis step

$$h = \frac{2(m-1)}{s}, \qquad m = 1 + \frac{hs}{2}. \tag{1}$$

For fixed $m = 2$, this step falls from 1 to $1/3$ between $\times 2$ and $\times 6$. A declining perturbation cost consequently combines changing input step with changing network response.

For native prediction P, perturbed prediction P_h, target T, native error $e = T - P$ and output change $\delta = P_h - P$, the MSE change is

$$D = \|e - \delta\|^2 - \|e\|^2 = \underbrace{\|\delta\|^2}_{J} - 2\,\mathrm{Re}\langle e, \delta\rangle. \tag{2}$$

We report J, D and the alignment term in RGB-mean MSE units. Small cost can arise from an output change that corrects existing error. The dB cost additionally depends on native MSE V through $10\log_{10}(1 + D/V)$.

Prospective equal-step test. Protocol, predictions, input identities and scoring code are frozen before inference on all 246 images in DIV2K 0869–0900, Set14, BSD100 and Urban100, using LIIF/LTE/SC-INR with two seeds. Common top-left modcrop-12 fixes HR support across scales; inputs use anti-aliased bicubic resizing, outputs remain unclipped, and measurements shave four pixels. We set $h \in \{-0.25, 0.25, 0.5\}$ on both axes. Complete r and u spectra, including DC and Nyquist, sum to the spatial metrics.

The primary prediction is a material violation of universal monotone response decay: at least one of the 36 prespecified dataset–step–adjacent-scale ratios has a one-sided, Bonferroni-corrected bootstrap lower bound above 1.10. We use 10,000 paired image resamples per dataset after averaging the fixed architectures and seeds. Separate predictions require a supported reversal in all 12 dataset–step conditions, and Set14 $J_6/J_2 > 1$ at both positive steps with corrected lower bounds above one. These gates are scored separately.

Calibrated responses remain substantial. The primary gate passes: 11/36 adjacent comparisons have family-wide corrected increases exceeding 10%, including seven positive-step $\times 2 \to \times 3$ comparisons with cells inside the training interval. All 12 dataset–step conditions pass their separate three-comparison gates (Fig. 2). The auxiliary Set14 endpoint prediction fails: at $h = 0.25$, $J_6/J_2 = 1.65$ with interval $[0.96, 2.71]$.

Response and cost can move in opposite directions. On BSD100 at $h = 0.5$, $J_6/J_2 = 1.24$ $[1.06, 1.44]$, but D falls from 79.42 to 40.77 in 10^{-6} MSE (paired change -38.65 $[-55.13, -22.01]$; descriptive 95% intervals, without multiplicity correction). Across datasets at this step, $r \geq 0.375$ accounts for 43.5–50.7% of J at $\times 2$, but only 2.4–3.0% at $\times 6$;

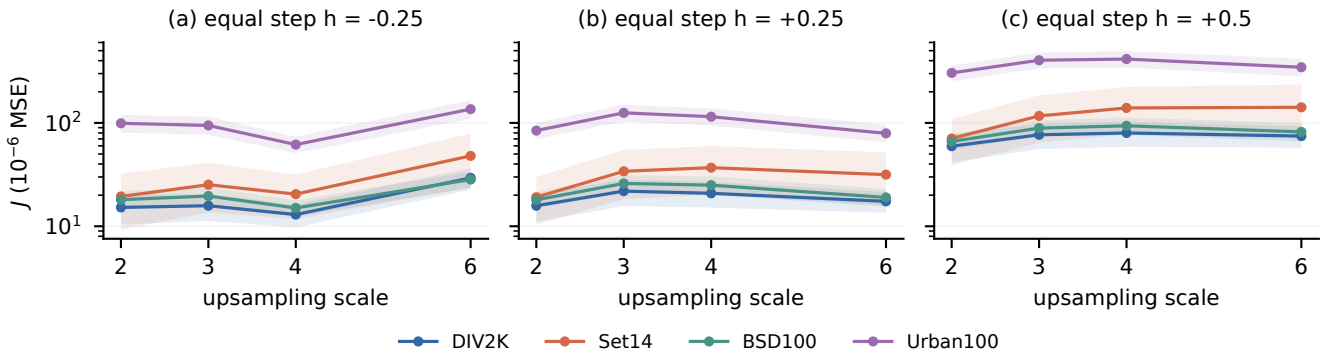


Fig. 2. Equal-step output responses need not decrease with scale. Each curve averages three decoder families and two seeds on one full-image dataset. Shading shows pointwise 95% paired-image bootstrap intervals, conditional on these checkpoints; the logarithmic axis reports J. Prediction gates use separately specified multiplicity corrections.

Table 1. Mean full-image Y-PSNR change (dB) relative to native EDSR-LIIF: four datasets, $\times 2/3/4/6$, matched 200-epoch arms and two seeds. Shuffled cell uses native cell at evaluation. †One seed. Parameter counts and compute differ for capacity changes.

Change	$\times 2$	$\times 3$	$\times 4$	$\times 6$
MLP 2×64	-0.25	-0.26	-0.28	-0.24
no cell input	-0.13	-0.09	-0.11	-0.10
constant cell		pending		
shuffled cell		pending		
RDN encoder	$+0.35$	$+0.29$	$+0.28$	$+0.25$
SwinIR encoder†	$+0.46$	$+0.42$	$+0.42$	$+0.34$

the middle band $r \in [0.1875, 0.375)$ accounts for 60.5–66.7% at $\times 6$.

These finite interventions compare responses across changing features and native conditions. Negative steps leave the training cell interval at some scales, and native $\times 6$ is outside it; marginal support is reported for each condition.

Scope of a scalar explanation. A retrospective real-gain fit gives $Q = \sum_b (1 - w_b)^2 R_b E_b$, fitting w_b to native and perturbed outputs. In the original DIV2K bands, doubled-cell Q/D is 0.44, 0.72, 0.81 and 2.50 at $\times 2/3/4/6$. Recovery and fitted-filter changes jointly account for 41.4% of the $\times 2 \to \times 6$ MSE-cost decline (Set14: 41.5%); alignment and non-scalar terms complete the accounting. Projections, image rankings and negative cases remain in the repository.

4. WHAT CELL CONTRIBUTES DURING LEARNING

Removing cell during training loses approximately 0.1 dB across benchmark scales in the existing two-seed comparison. To test the source of this benefit, we add two EDSR-LIIF arms under the same 200-epoch recipe and two seeds. Both retain the native decoder's two cell inputs and parameter count. Constant-input weights can be absorbed into the first-layer bias, adding no expressivity. The *constant* arm receives the training mean of the actual cell coordinate, $c_0 = \mathbb{E}[96/\,\mathrm{round}(48s)]$ for $s \sim \mathcal{U}(1, 4)$, at every query and evaluation geometry. The *shuffled* arm receives another sample's complete cell vector, using a random batch derangement. This preserves the empirical cell marginal while removing the correct pairing. A separate random generator preserves the data and model-initialisation streams, and assignment statistics verify the intervention. Queries and image features can still carry other scale cues.

We freeze predictions and scoring before training, select epoch 200 regardless of validation, and reevaluate the existing native/no-cell checkpoints with the same evaluator. Standard full-image Y-PSNR follows the original PIL-bicubic protocol, clamps outputs and shaves by scale on Set5, Set14, BSD100 and Urban100. We also measure quarter-u recovery on the original DIV2K crop cohort. The shuffled arm's primary evaluation uses native cell; fixed-mean and four independent marginal-cell assignments per image provide predeclared sensitivity checks.

The primary predictions require native cell to outperform each new arm by at least 0.05 dB in the dataset- and scale-balanced mean, with both seeds positive and a Bonferroni-adjusted 97.5% paired-image interval above zero. Separate equivalence tests ask whether each arm lies within ± 0.03 dB of no-cell; failure to establish equivalence is retained. Training comparisons use 2,000 within-dataset paired image resamples, conditional on the two seeds.

Review-draft status. All four frozen 200-epoch runs are in progress. Benchmark effects, recovery localisation, inference-policy sensitivity and the four prediction verdicts will be reported after the integrated pipeline completes; no partial validation result is used as a final outcome. These contrasts assess correct pairing, retained dimensions and training noise; their interpretation is limited to this recipe.

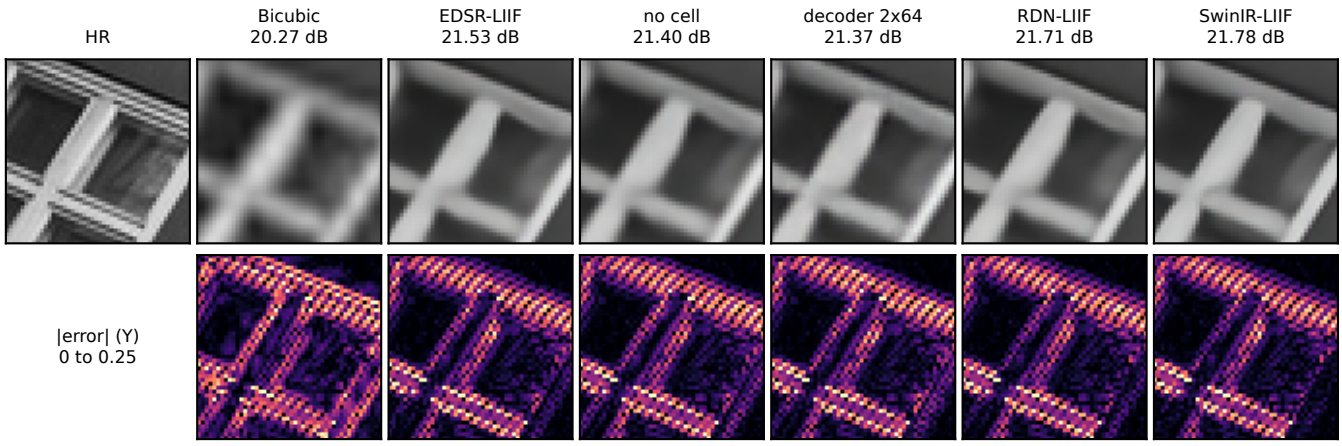


Fig. 3. Rule-selected $\times 4$ recovery example, Urban100 img_036. Before visual inspection, the image is selected at the upper-median rank of whole-image RDN–EDSR Y-PSNR gain (51st of 100); the 48×48 window maximises HR gradient energy on a stride-eight grid. Top: reconstructions and window Y-PSNR. Bottom: absolute Y error, shared range $[0, 0.25]$. The fixed example illustrates the existing capacity and no-cell arms.

5. CONCLUSION AND SCOPE

Recovery profiles locate a shared range in which trained capacity improves reconstruction. Cell analysis requires a calibrated decoder-input step and separate measurements of output change and error alignment. Dimension-preserving training controls then test the value of correctly paired conditioning. Equal-step measurements reject universal response decay while retaining a failed endpoint prediction. The source of the training benefit remains open until the registered comparisons complete.

Evidence is limited to the tested architectures, recipes and two degradations. New measurements were specified prospectively on previously studied images/checkpoints. Image-bootstrap intervals condition on fixed trained seeds; finite interventions and ablations do not identify a unique internal mechanism.

Table 2. Registered predictions and retained failures. Counts are satisfying checks, followed by overall pass (P) or fail (F); they are not independent experiments. All original prediction records remain in the repository. Stages 17–18 are frozen prospectively; the scalar decomposition is retrospective.

Prediction	Outcome
strict $R(u)$ collapse (crops; full)	45/54; 53/72 F
decoder spread ≤ 0.03 (crops; full)	36/36; 54/54 P
doubled-cell peak $r \in [.19, .375)$	36/36 P
fixed-multiplier cost declines	9/9 P
$Q/D \in [.7, 1.3]$ at $\times 3/4$	6/6 P
best cell within 10% of native ($s \leq 4$)	20/27 F
smaller cell costs more	33/144 F
small MLP harmless at large scales	F
blur moves gains below $u = 0.75$	0/3 F
$\geq 80\%$ absolute cost lies at $u \geq 1$	311/414 F
external RDN-BTC; EDSR gain	0/12; 0/4 F
external cell locus (GaussianSR; BTC)	2/4; 3/4 F
no-cell loss decays to negligible cost	F
17: corrected adjacent-scale increase	11/36 P
17: increase in all 12 conditions	12/12 P
17: both positive-step Set14 endpoints	1/2 F
18: native advantage (constant; shuffle)	pending
18: no-cell equivalence (constant; shuffle)	pending

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